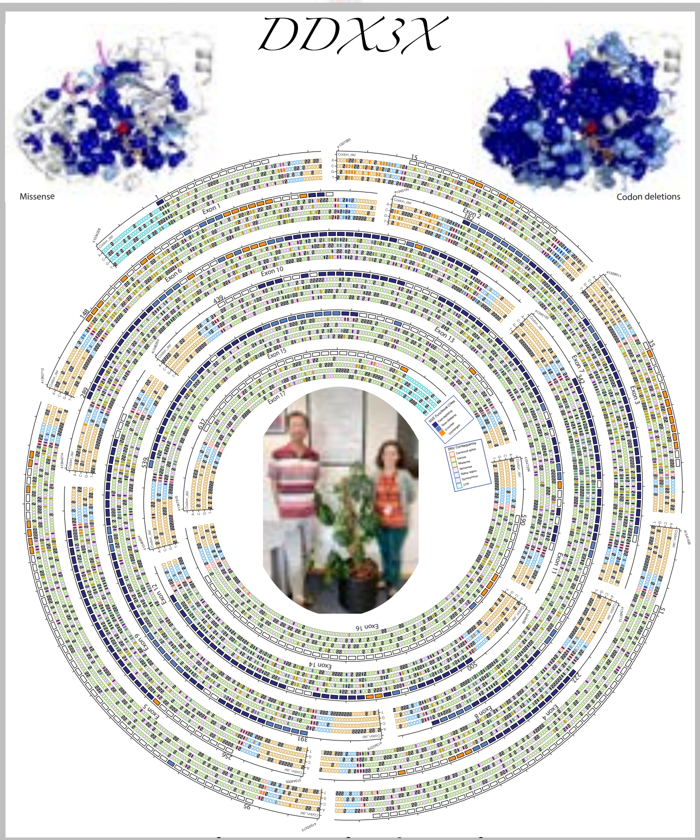
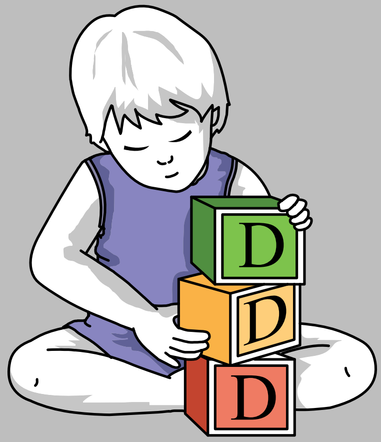
Saturation Genome Editing: Understanding variant effect at scale in an endogenous genomic context





Sebastian Gerety, Principal Staff Scientist
sg13@sanger.ac.uk
Hurles group
Human Genetics, Wellcome Sanger Institute

Variant Effects In The Context Of Rare Developmental Disorders

Matt Hurles' group works on the genetic basis of developmental disorders (DDs)

- **Individually rare, but together** affect 3-5% of the population
- Strong genetic component
- Early onset
- dramatic impact on individuals, families, health care system
- High incidence of cognitive, neurological, and behavioural impairment
- Often include metabolic deficits, growth deficiencies, and congenital anomalies

Large scale sequencing projects identify new disease-gene associations

DDD Project: Deciphering Developmental Disorders

Exome sequencing of >30,000 Family trios with an affected child

- **Research**: Novel disease gene identification
- **Diagnostics**: Feedback likely genetic diagnoses

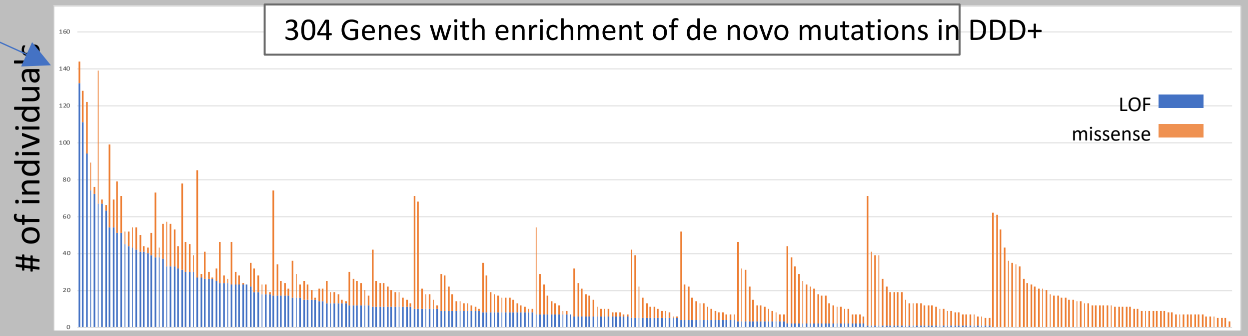
>300 genes with excess frequency of *de novo* mutation in the cohort

Heterozygous missense and protein truncating variants

Discovery: >50 previously unidentified DD genes found

Burden: ~50% of probands likely to carry causal *de novo* mutation

140



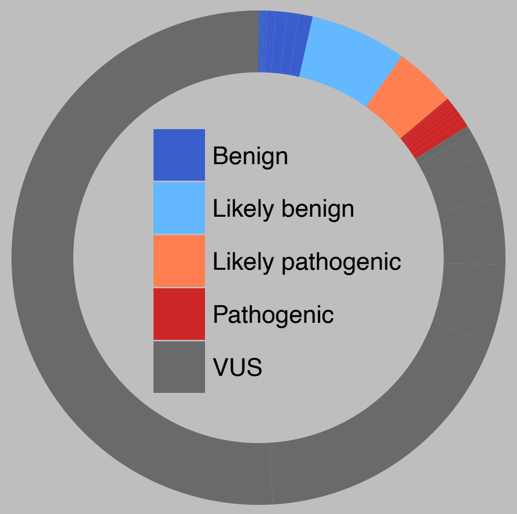
30K Collaboration: NHS - Sanger Institute - GeneDX - Radboud University
Kaplanis J, Samocha KE, et al Nature. 2020 Oct;586(7831)

Number of variants identified in patients is increasing exponentially

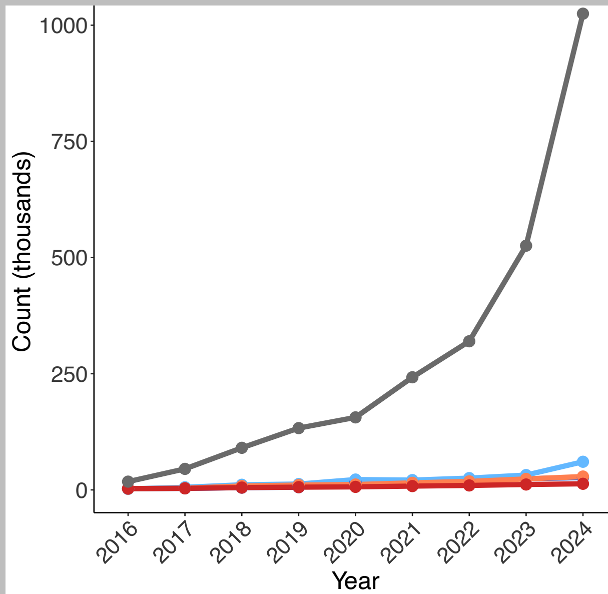
For most of these, we do not know whether they impact protein function:

Variants of Uncertain Significance: VUS

- typically missense
- uncertain impact on protein function
- prevent definitive diagnosis for patients (don't meet diagnostic criteria)
    - limit available treatment
    - delay timely intervention
    - isolate affected individuals and their families from communities and knowledge



Fayer S. et al. 2024



Variant Effects In The Context Of Rare Developmental Disorders

Number of variants identified in patients is increasing exponentially

For most of these, we do not know whether they impact protein function:

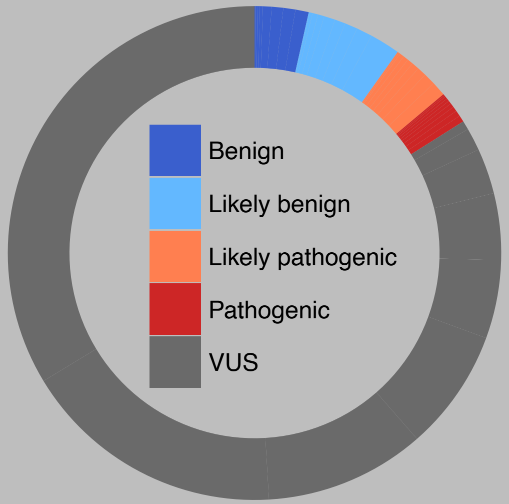
Variants of Uncertain Significance: VUS

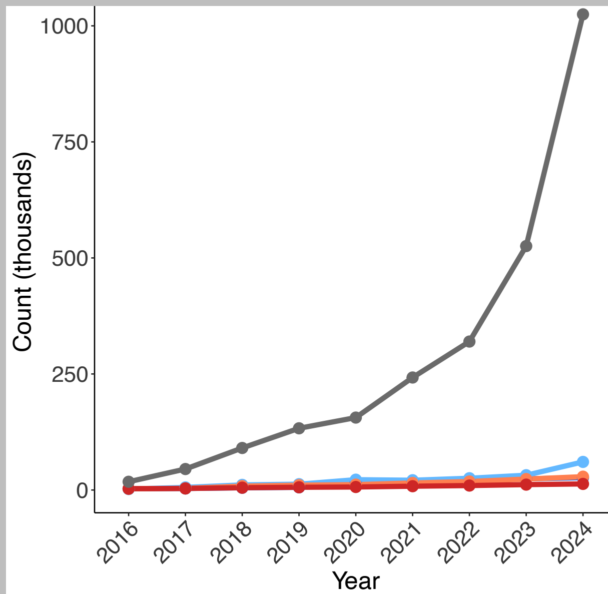
- typically missense
- uncertain impact on protein function
- prevent definitive diagnosis for patients
    - limit available treatment
    - delay timely intervention
    - isolate affected individuals and their families from knowledge

RESOLVED VUS

PROVIDES DIAGNOSIS
INFORMATION AND ACCESS TO EXISTING TREATMENT
ENABLES EARLY INTERVENTION
PROVIDES COMMUNITY, KNOWLEDGE
PROVIDES PROGNOSTICS ON DISEASE PROGRESSION

STRONG MOTIVATION TO RESOLVE VARIANTS OF UNCERTAIN SIGNIFICANCE





Fayer S. et al. 2024

To capitalise on bounty of diagnostic sequencing we must resolve more VUS

- Computational effect predictors are not yet perfected
- Experimental evidence is accepted diagnostic evidence in clinical genetics

Assays to test variant effect

- previously low throughput, high cost often gene-specific, e.g.
  - performed on a small number of variants of interest
  - protein stability assays
  - protein activity (phosphorylation, DNA binding, transcriptional readouts, localisation)

Variant Effects In The Context Of Rare Developmental Disorders

To capitalise on bounty of diagnostic sequencing we must resolve more VUS

- Computational effect predictors are not yet perfected
- Experimental evidence is accepted in clinical genetics

Assays to test variant effect

- previously low throughput, high cost often gene-specific, e.g.
  - Performed on a small number of variants of interest
  - Protein stability assays
  - Protein activity (phosphorylation, DNA binding, transcriptional readouts, localisation)

Current solution, **Supersized variant assays**

- **MAVE: Multiplex Assays of Variants Effect**
  1. Identify a readout or feature of protein inactivation/perturbation (novel or known)
  2. Choose method to generate variant-carrying proteins
  3. Establish/apply assay to test many variants together
  - Allows for scaling up
    - increased efficiency -> feasibility of doing whole genes +
    - cost -> per variant costs reduced

MAVE: Multiplex Assays of Variants Effect - Many ways to skin a cat

No assay is perfect – adjust your choice to your needs

QUESTION	OPTIONS	PROS	CONS
Coding vs non-coding variation	Some assays tailored to regulatory sequences, others to coding, others agnostic		
Protein-coding MAVES			
What do you assay	SGE Full protein	Asays impact on whole protein	Longer constructs/cargo More expensive
	Domains, fragments	Easier, cheaper to construct libraries	Results less likely to be relevant to whole protein variant effect
How do you introduce variants	Pre-generated library (Viral, recombination, HDR)	Unbiased methods of library generation by cloning	Higher cost, more upfront effort to make libraries
	Editing reagent (Prime editing, Base Editing)	Multi-gene and multi-locus assessment is possible	*see individual methods below*
How is variant protein expressed	Viral delivery of cDNA	High efficiency (selection for infected cells) Same viral pool can be assayed in many cell-types	Insertion locus affects expression level ->increased variation = reduced sensitivity Cargo size limits dependent on virus High false negative rate: variants that block splicing, induce splicing are missed Regulatory variants not accessible Nonsense mediated decay completely masked (no introns) Non-pysiological expression levels
	Recombination (e.g. Landing pad) delivery of cDNA	High efficiency, depending on cell line Homogeneous expression from landingpad locus Same reagent can be assayed in multiple celltypes (LP inserted)	Cargo size limits dependent on delivery High false negative rate: variants that block splicing, induce splicing not captured Regulatory variants not accessible Nonsense mediated decay completely masked: no introns
	SGE Endogenous editing - CAS9+HDR (Saturation Genome Editing)	Genomic structure intact Intron structure retained -> Nonsense mediated decay preserved Splice sites preserved->splice site variants can be tested Splicing is preserved -> novel splice site variants can de detected Nonsense Mediated Decay is active, effects are captured	Can be limited to specific locus, protein, exon Whole proteins often need splitting into exon-sized units: higher costs (more experiments)
	Endogenous editing - Prime Editing	As above + Genome-wide editing within one experiment is possible	Efficiency is typically poor in most cell lines Efficiency of reagents not currently predictable
	Endogenous editing - Base Editing		Very limited edits available from Base editors Limited precision of editing (shotgun)
What platform is used	SGE Mammalian cells	Highest species relevance: splicing, post-translational modifications, NMD, regulation, binding partners, target genes	More costly culture reagents, larger cells = lower coverage
	Yeast	Eukaryotic, but cheaper culture reagents	Less relevant to human biology and disease, post-translational modifications, NMD, regulation, binding partners, target genes
	Bacteria	Extremely inexpensive experiments, massive scale/coverage	No eukaryotic biology captured: post-translational modifications, NMD, regulation, binding partners, target genes
What readout is used	Protein-specific	Tailored to specific protein under study	Disease relevance not always clear Set up time is long, increasing cost Assay not reusable, increasing cost
	SGE Generic-cell growth	Can multiplex multiple known readouts for complex readout	Limited to 1 dimensional readout (growth rate)
	Generic-transcriptional readout (scRNAseq)		Requires known growth impact of variants
	Generic - cell morphology, cell painting		Disease relevance not always clear Not yet robustly established, demonstrated to work

Features of Saturation Genome Editing: SGE

Accurate classification of *BRCA1* variants with saturation genome editing

Gregory M. Findlay[1], Riza M. Daza[1], Beth Martin[1], Melissa D. Zhang[1], Anh P. Leith[1], Molly Gasperini[1], Joseph D. Janizek[1], Xingfan Huang[1], Lea M. Starita[1,2*] & Jay Shendure[1,2,3*]

Saturation Genome Editing

- Established by Greg Findlay in the Shendure Group in 2018

1. Creating variant proteins

- Uses **direct genome editing** to introduce variants into endogenous gene locus: **Homology Directed Repair - CAS9 enhanced HDR**
  - Preserves intact gene locus, including all regulatory and splice features, and physiological expression levels
  - Captures important effects impacting disease mechanism
    - loss of splice sites
    - gain of novel splice sites
    - impact of nonsense mediated decay on stability of variant mRNA

2. Assay to asses impact

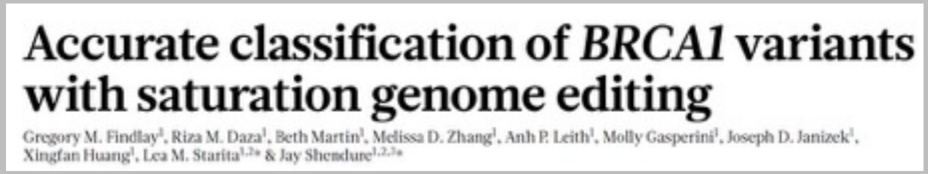
- relies on growth assay
  - **Gene of interest must be essential in cell-type being used (HAP1 cells)**
  - Variants that damage protein function lead to poor cell growth
  - Cells carrying damaging variants therefore grow more slowly
  - Over time, proportion of cells carrying a damaging variant will drop as unaffected cells continue to proliferate
  - Assaying cell population at different timepoints by sequencing edited locus reveals depletion of deleterious variants
    - dropout screen

3. Experimental setup to test regions at scale

- Each experiment targets small region of genome
  - typical targeted region is 1 exon: <300 bases
- Entire gene can be assayed through numerous small experiments
- "Typical" targeted gene is 23 exons: 23 experiments

SGE: Saturation Genome Editing

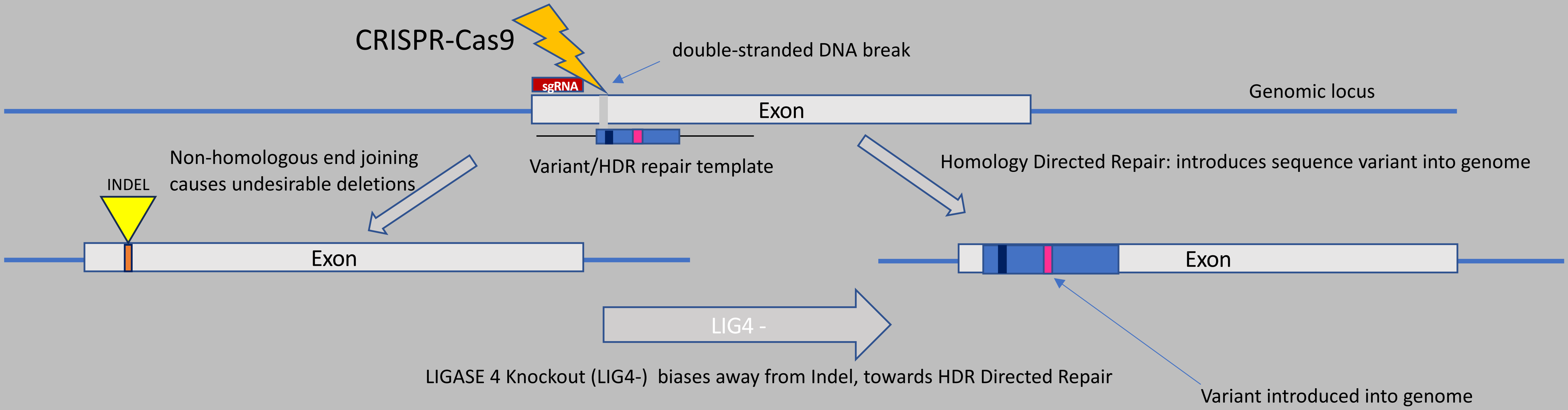
Saturation Genome Editing
- Established by Greg Findlay in the Shendure Group in 2018


Accurate classification of *BRCA1* variants with saturation genome editing

Gregory M. Findlay[1], Riza M. Daza[1], Beth Martin[1], Melissa D. Zhang[1], Anh P. Leith[1], Molly Gasperini[1], Joseph D. Janizek[1], Xingfan Huang[1], Lea M. Starita[1,2*] & Jay Shendure[1,2,3*]

1. Creating variant proteins
- Uses **direct genome editing** to introduce variants into endogenous gene locus: **Homology Directed Repair - CAS9 enhanced HDR**
  - Preserves intact gene locus, including all regulatory and splice features, and physiological expression levels
  - Captures important effects impacting disease mechanism
    - loss of splice sites
    - gain of novel splice sites
    - impact of nonsense mediated decay on stability of variant mRNA

HOMOLOGY DIRECTED REPAIR



SGE: Saturation Genome Editing

Saturation Genome Editing

- Established by Greg Findlay in the Shendure Group in 2018

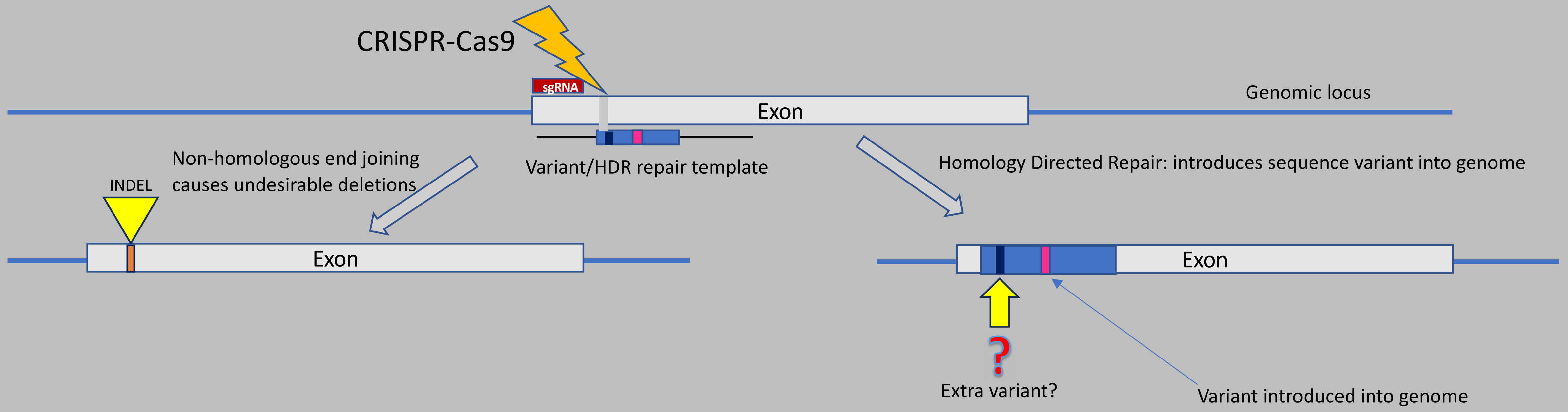
Accurate classification of *BRCA1* variants with saturation genome editing

Gregory M. Findlay[1], Riza M. Daza[1], Beth Martin[1], Melissa D. Zhang[1], Anh P. Leith[1], Molly Gasperini[1], Joseph D. Janizek[1], Xingfan Huang[1], Lea M. Starita[1,2*] & Jay Shendure[1,2,3*]

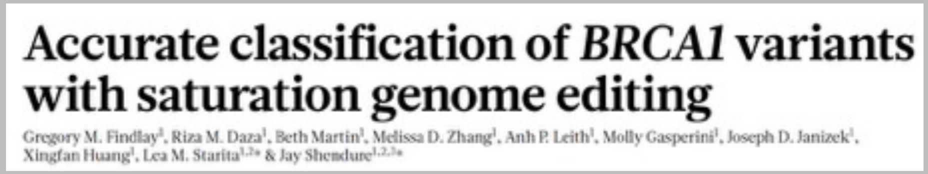
1. Creating variant proteins

- Uses **direct genome editing** to introduce variants into endogenous gene locus: **Homology Directed Repair - CAS9 enhanced HDR**
  - Preserves intact gene locus, including all regulatory and splice features, and physiological expression levels
  - Captures important effects impacting disease mechanism
    - loss of splice sites
    - gain of novel splice sites
    - impact of nonsense mediated decay on stability of variant mRNA

HOMOLOGY DIRECTED REPAIR



SGE: Saturation Genome Editing


Accurate classification of *BRCA1* variants with saturation genome editing

Gregory M. Findlay[1], Riza M. Daza[1], Beth Martin[1], Melissa D. Zhang[1], Anh P. Leith[1], Molly Gasperini[1], Joseph D. Janizek[1], Xingfan Huang[1], Lea M. Starita[1,2*] & Jay Shendure[1,2,3*]

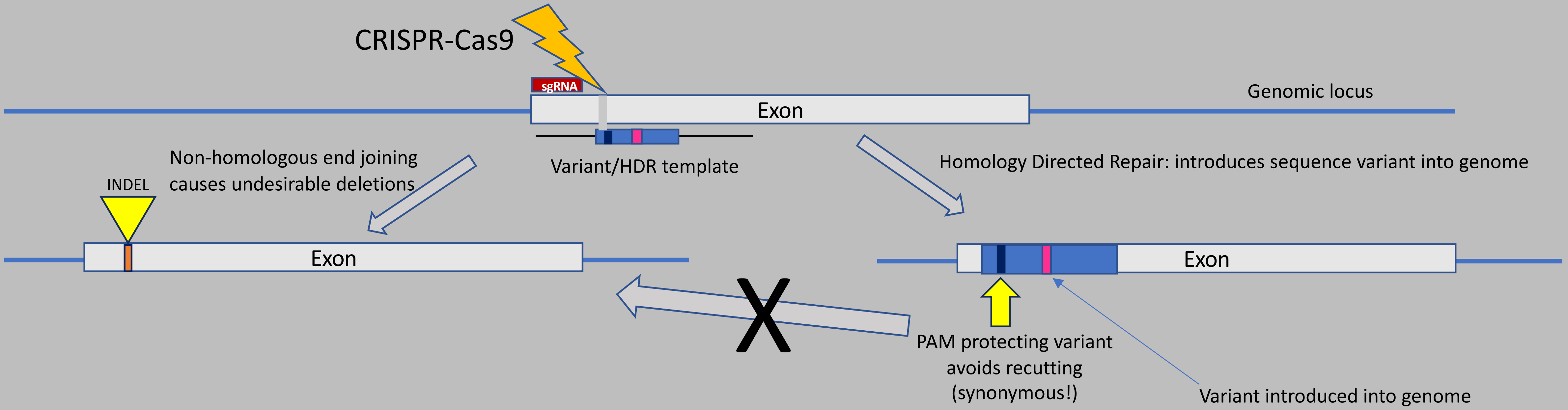
Saturation Genome Editing

- Established by Greg Findlay in the Shendure Group in 2018

1. Creating variant proteins

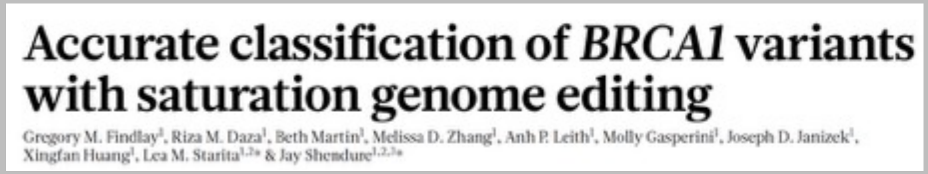
- Uses **direct genome editing** to introduce variants into endogenous gene locus: **Homology Directed Repair - CAS9 enhanced HDR**
  - Preserves intact gene locus, including all regulatory and splice features, and physiological expression levels
  - Captures important effects impacting disease mechanism
    - loss of splice sites
    - gain of novel splice sites
    - impact of nonsense mediated decay on stability of variant mRNA

HOMOLOGY DIRECTED REPAIR



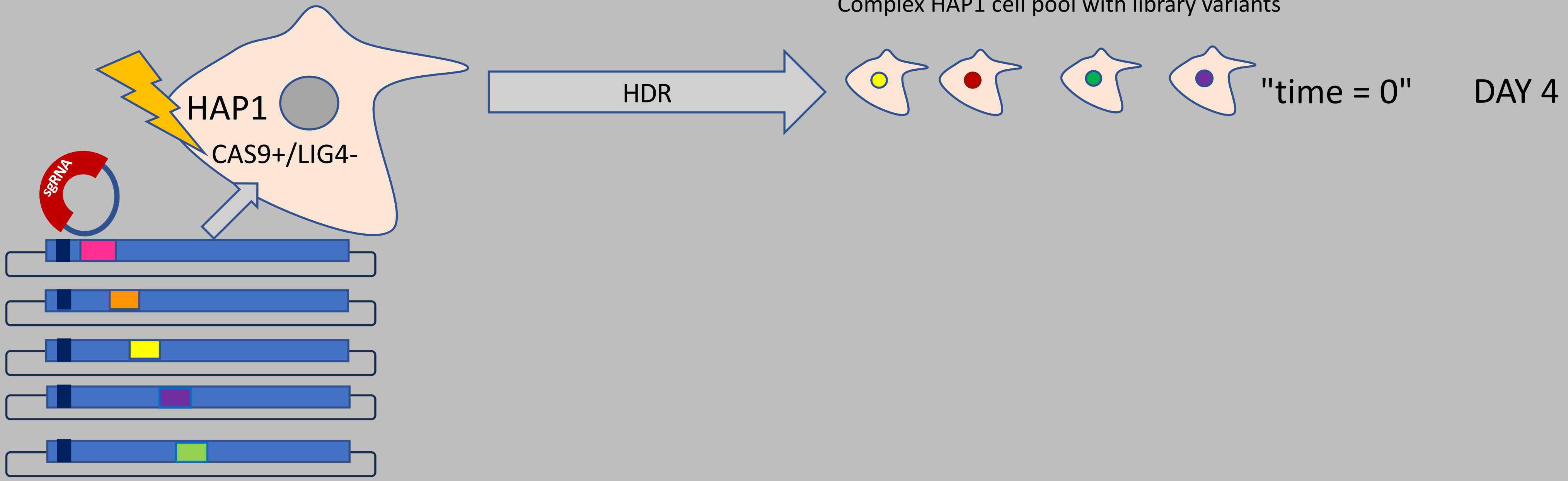
SGE: Saturation Genome Editing

Saturation Genome Editing
- Established by Greg Findlay in the Shendure Group in 2018


Accurate classification of *BRCA1* variants with saturation genome editing

Gregory M. Findlay[1], Riza M. Daza[1], Beth Martin[1], Melissa D. Zhang[1], Anh P. Leith[1], Molly Gasperini[1], Joseph D. Janizek[1], Xingfan Huang[1], Lea M. Starita[1,2*] & Jay Shendure[1,2,3*]

1. Creating MANY variant proteins
- Using a **library of HDR templates carrying many variants**, we generate a population of cells with full coverage of library
  - HAP1 cells are **haploid**, so carry only one copy of each gene/locus
    - Each cell expresses only one variant protein
    - Any impact observed is due to the one variant a cell carries



Variant library

SGE: Saturation Genome Editing

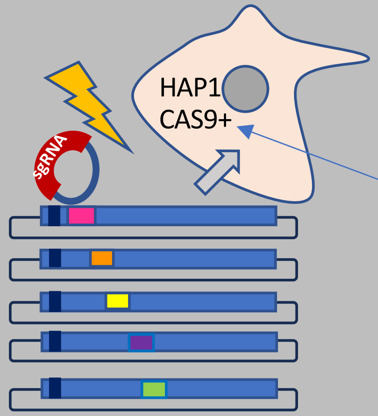
Saturation Genome Editing: Sausage-making slide

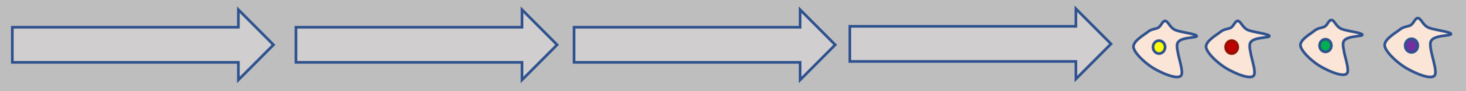
1. Creating variant proteins

DAY 0
Transfection

DAYS 1,2,3
PUROMYCIN SELECTION
ENRICHES FOR MOST HIGHLY TRANSFECTED CELLS
Selects for cells likely to undergo HDR

DAY 4/5
Cell pellet collected for "time = 0"
remainder passaged for continued growth





Stably integrated CAS9 expression cassette
Provides strong, reproducible CAS9 activity and HDR
(transient CAS9 works, but less well)

SIZE DOES MATTER
Larger plasmids transfect less efficiently
Removing the CAS9 ORF from our plasmids:
- increased transfection efficiency
- raised HDR rates
- reduced wildtype fraction to zero

Variant library

DAY 0
Transfection

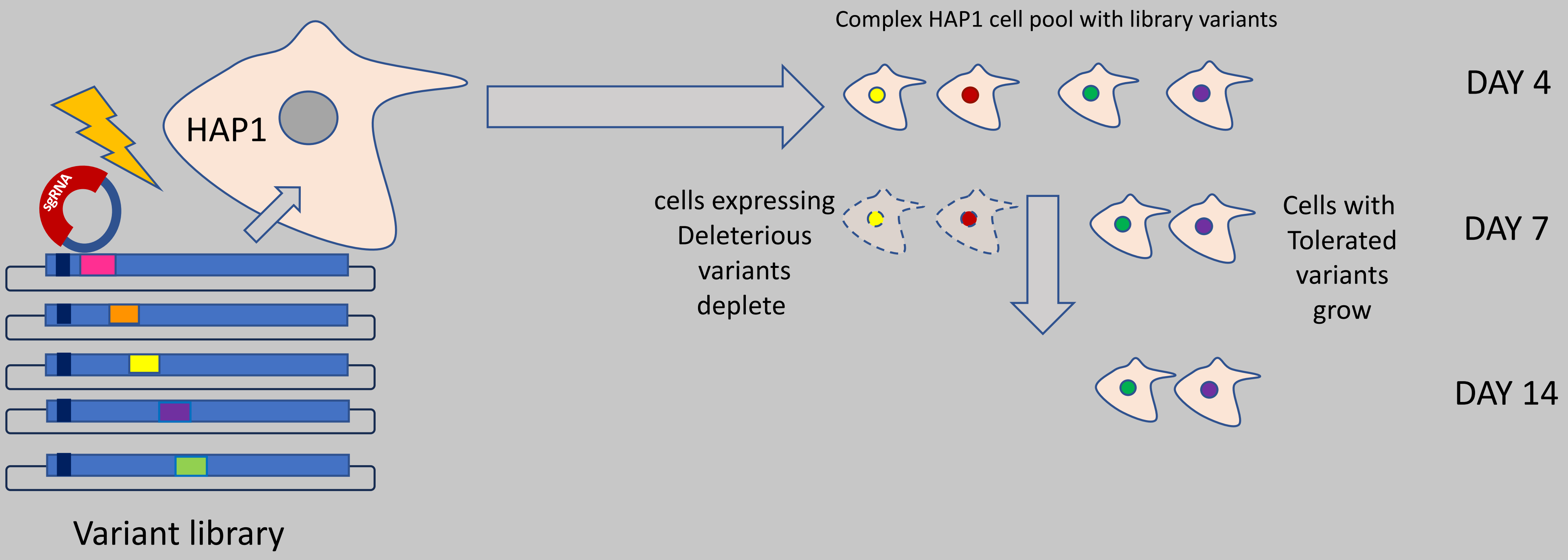
sgRNA plasmid with PURO selection cassette
+
HDR template library plasmid pool
(100—2000 variants)

SGE: Saturation Genome Editing

Saturation Genome Editing

2. Assay to asses impact

- relies on growth assay
  - gene of interest **must be essential in cell-type being used** (HAP1 cells)
  - variants that damage protein function lead to poor cell growth
  - cells carrying damaging variants therefore grow more slowly
  - over time, **proportion of cells carrying a damaging variant will drop** as unaffected cells continue to proliferate

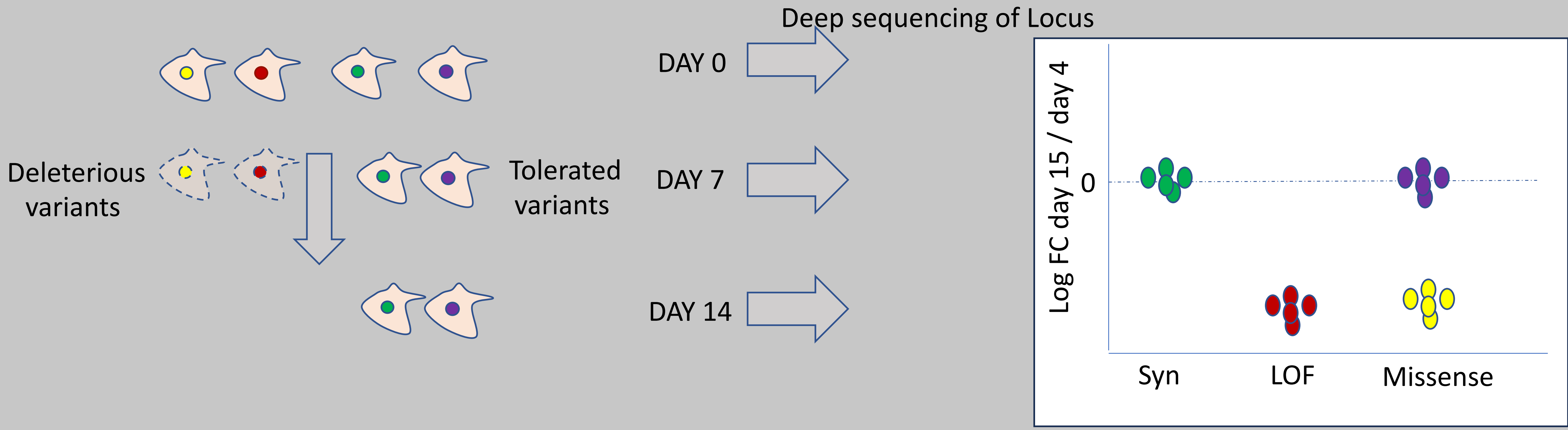


SGE: Saturation Genome Editing

Saturation Genome Editing

2. Assay to asses impact

- NextGen sequencing of PCR amplicons of edited locus
- provides reads for variant counting
- counts = cells carrying variants
- Change in representation relative to starting timepoint gives depletion metric: Log_2Fold change
- Many analysis tools, including DESEQ2



SGE: Saturation Genome Editing

REQUIREMENTS FOR SGE

1. **Gene of interest MUST BE ESSENTIAL IN HAP1**
2. **Mechanism under study MUST BE LOSS OF FUNCTION**

Then, variants of interest (LOSS OF FUNCTION) will lead to reduced cell growth effect

- This limits the scope of HAP1 based SGE:
  - Disease genes of LOF mechanism
  - Sufficient HAP1 essentiality

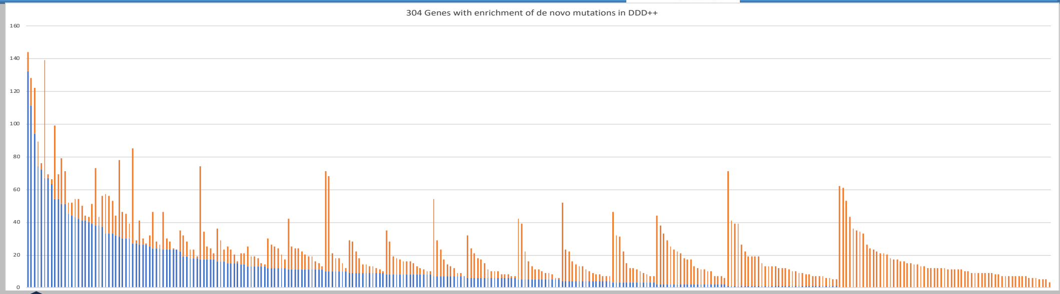
Applying SGE to Developmental Disorder Genes

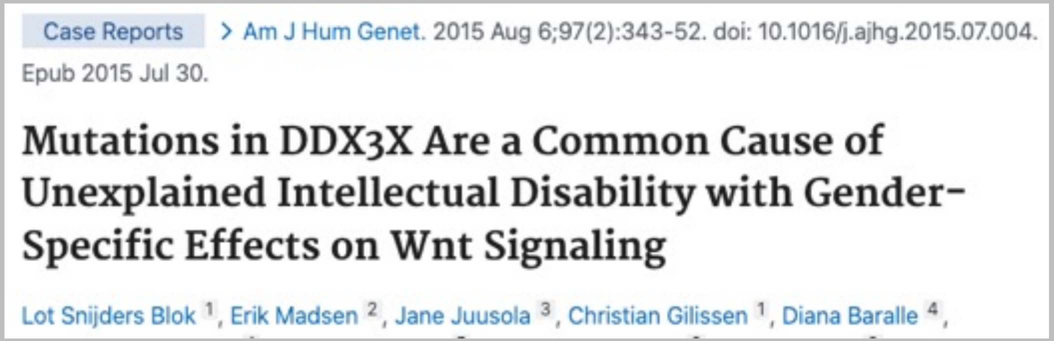
Hurles group set out to validate the use of SGE on NDD genes

DDX3X

- ✓ Very "high prevalence" disorder gene:
    - 120 LOF + missense in 30K proband DDD study
- X-linked
- Most common cause of ID in girls
    - Developmental delay or intellectual disability
    - Behaviour problems, including autism and ADHD
    - Low muscle tone (hypotonia)

- ✓ Caused by LOF variants
- ✓ Essential in HAP1




Case Reports > Am J Hum Genet. 2015 Aug 6;97(2):343-52. doi: 10.1016/j.ajhg.2015.07.004. Epub 2015 Jul 30.

Mutations in DDX3X Are a Common Cause of Unexplained Intellectual Disability with Gender-Specific Effects on Wnt Signaling

Lot Snijders Blok [1], Erik Madsen [2], Jane Juusola [3], Christian Gilissen [1], Diana Baralle [4],

Applying SGE to Developmental Disorder Genes

Hurles group set out to validate the use of SGE on NDD genes

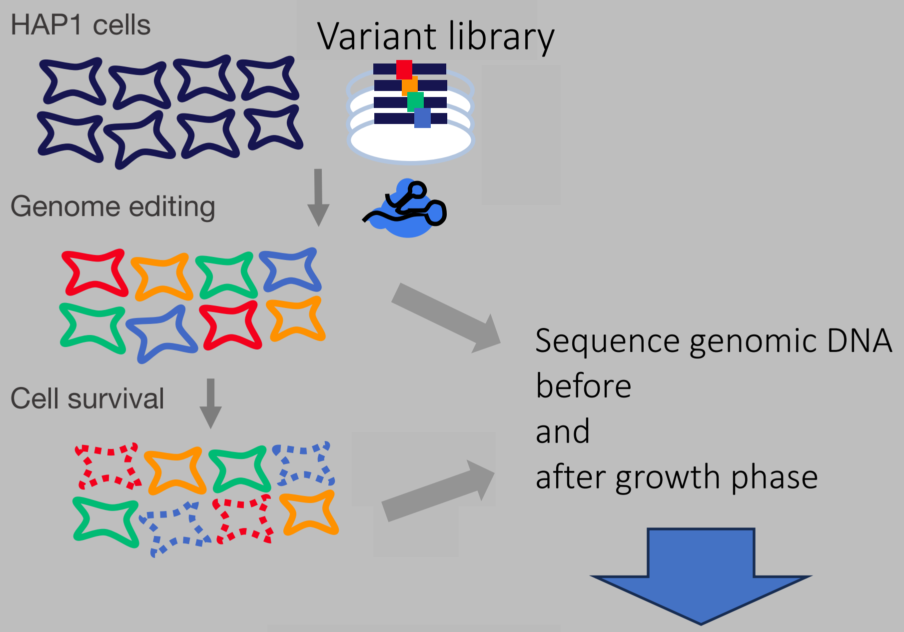
DDX3X

✓ Caused by LOF variants

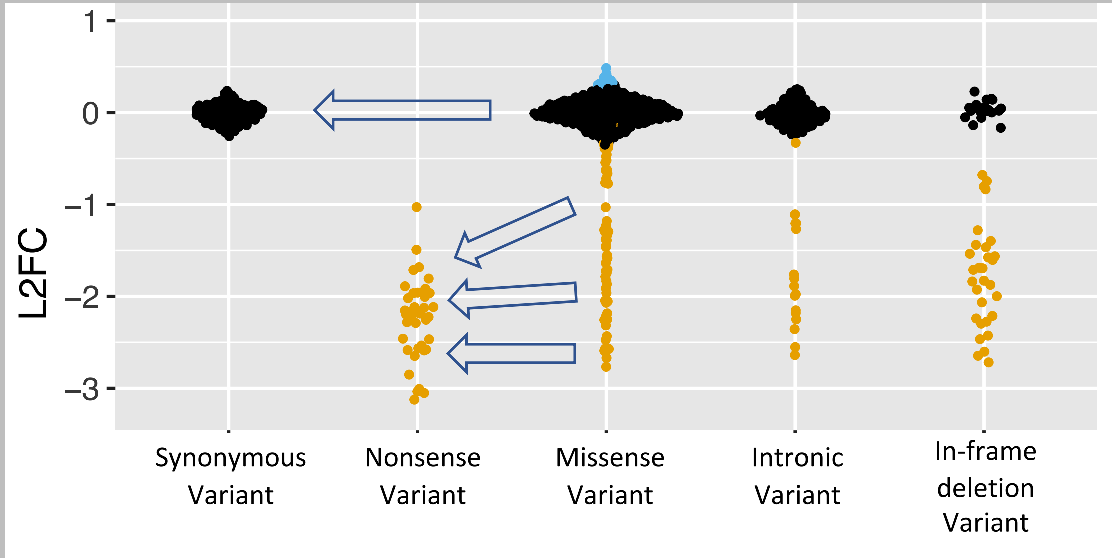
✓ Essential in HAP1

Variant behaviour at day 15 was as expected

- synonymous variant look unaffected (LFC=0)
- nonsense (LOF) variants all drop to lower LFC
- Missense variants display mix
  - some deplete: deleterious missense variants
  - some grow normally: benign variants
- Other variation shows varied impact

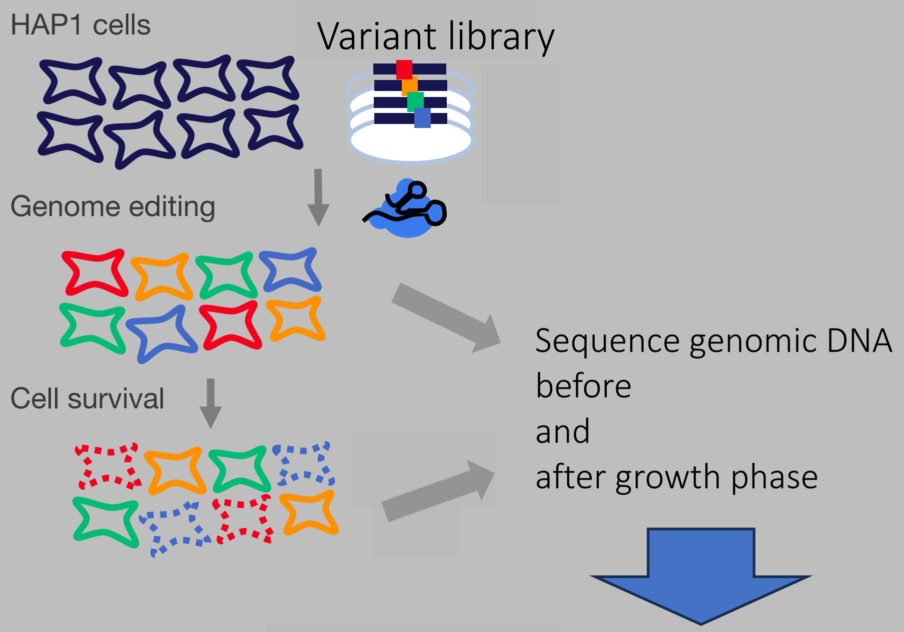


DDX3X Exon 15

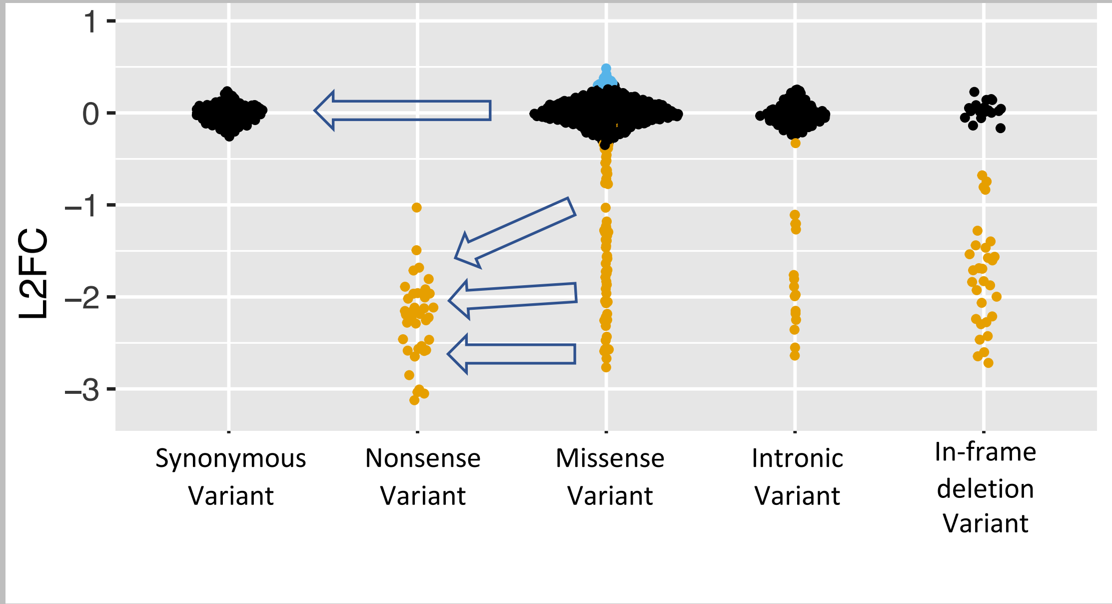


BEESWARM plots are your friend!

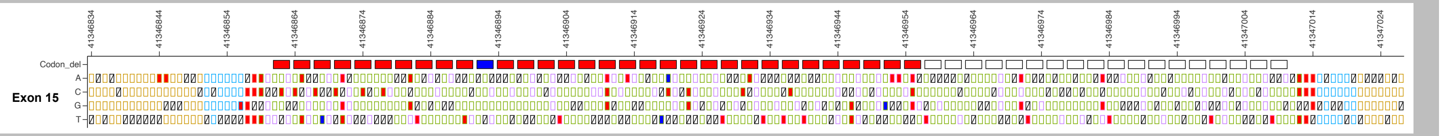
Applying SGE to Developmental Disorder Genes



DDX3X Exon 15



DDX3X Exon 15 Variant effect map



Applying SGE to Developmental Disorder Genes

DDX3X

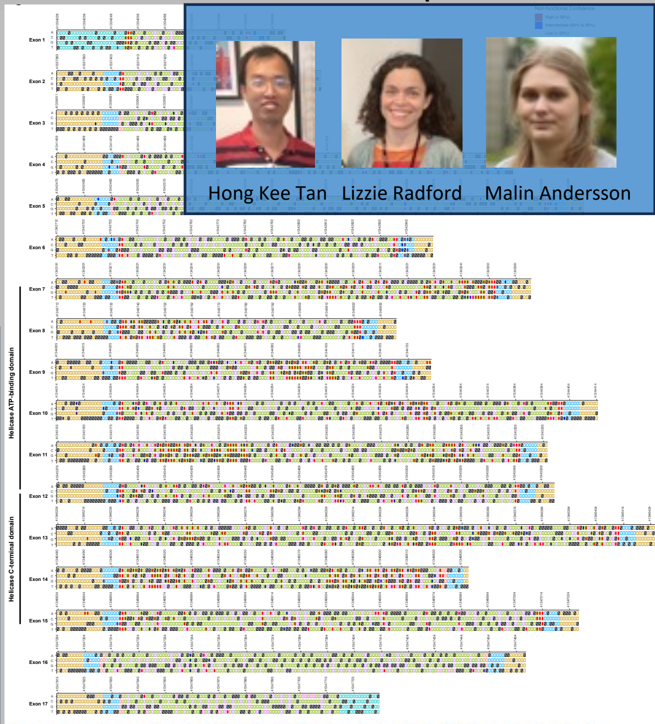
Whole gene map recapitulates observations in patients -> GOOD MODEL

- Model has 97% sensitivity and 99% specificity

Resolves up to 93% of variants of uncertain significance

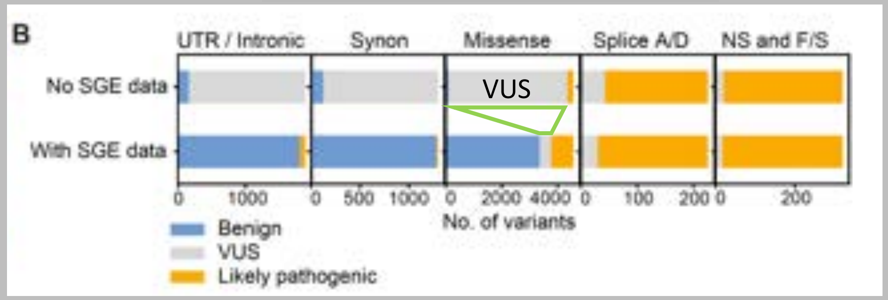
Can also detect likely somatic cancer driver mutations in DDX3X

DDX3X Variant effect map

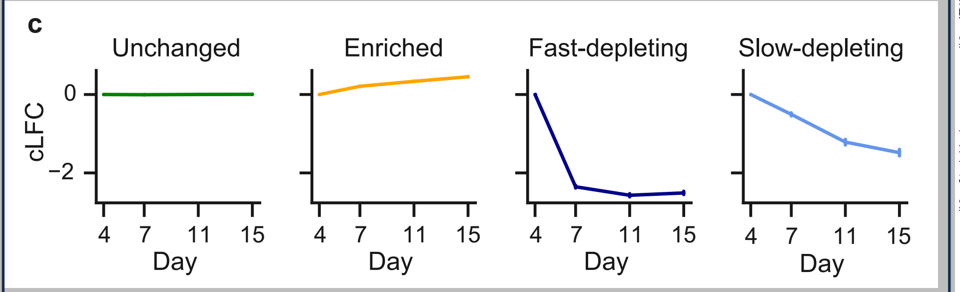


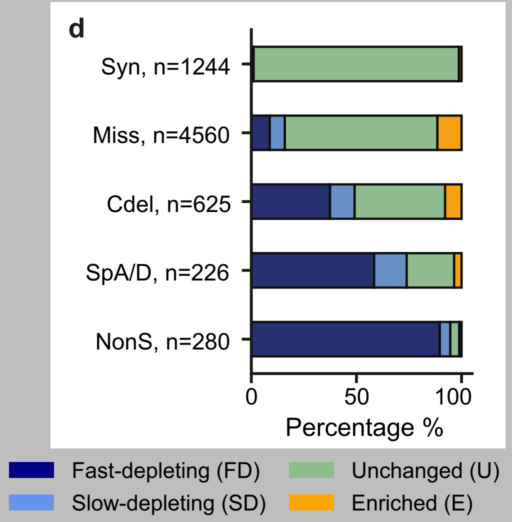
Saturation genome editing of DDX3X clarifies pathogenicity of germline and somatic variation.
Radford EJ, Tan HK, Andersson MHL, Stephenson JD, Gardner EJ, Ironfield H, Waters AJ, Gitterman D, Lindsay S, Abascal F, Martincorena I, Kolesnik-Taylor A, Ng-Cordell E, Firth HV, Baker K, Perry JRB, Adams DJ, **Gerety SS**, Hurles ME.
Nat Commun. 2023 Dec 6;14(1):7702. doi: 10.1038/s41467-023-43041-4.
PMID: 38057330 Free PMC article.

MAVE Resolves VUS

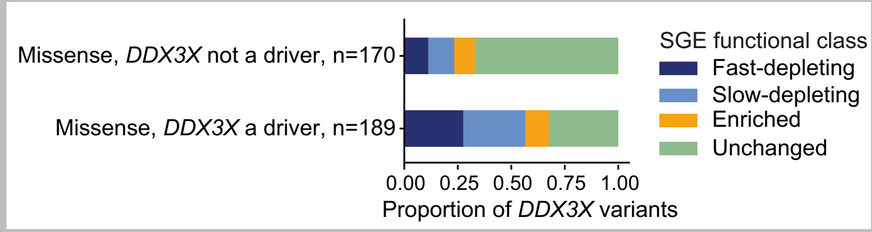


Variants differ in their depletion kinetics





Somatic DDX3X LOF variants enriched in certain cancers



Applying SGE to Developmental Disorder Genes

SGE in HAP1 is a valid and effective assay for variant effects in Neurodevelopmental disorders

As part of the global partnership - The AVE Alliance – we aim to generate and release increasing numbers of variant effect maps

Selecting genes

- Disease associated genes of interest
- LOF Mechanism
- **Essential in HAP1**

Critical factors in SGE success

Essentiality in HAP1

We generated a highly efficient haploid HAP1 LIG4^{KO} CAS9 expressing clonal line: High HDR rates

Distributed globally through Horizon (https://horizondiscovery.com/en/engineered-cell-lines/products/human-hap1-knockout-cell-lines?nodeid=entrezgene-3981)

While multiple publications listing HAP1 essential genes, having essentiality data in your own cell line is critical

Why does it matter?

- **Accurate essentiality scoring critical** for gene triage and experimental planning
  - Inaccurate scores, e.g. from a different cell line, increases failure rate
    - weakly essential genes may need longer screening times
    - too-weakly essential/non-essential genes will fail

> Science. 2015 Nov 27;350(6264):1092-6. doi: 10.1126/science.aac7557. Epub 2015 Oct 15.

Gene essentiality and synthetic lethality in haploid human cells

Vincent A Blomen [1], Peter Májek [2], Lucas T Jae [1], Johannes W Bigenzahn [2], Joppe Nieuwenhuis [1], Jacqueline Staring [1], Roberto Sacco [2], Ferdy R van Diemen [1], Nadine Olk [2], Alexey Stukalov [2], Caleb Marceau [3], Hans Janssen [1], Jan E Carette [3], Keiryn L Bennett [2], Jacques Colinge [4], Giulio Superti-Furga [5], Thijn R Brummelkamp [6]

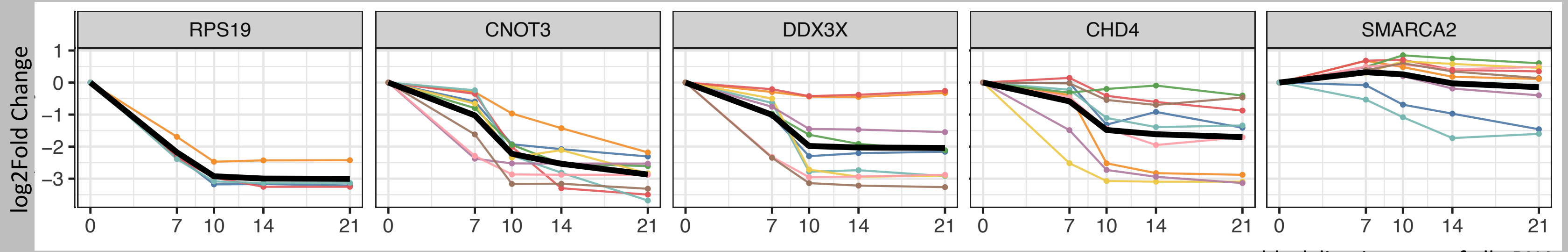
We (Sanger) recently completed a whole genome essentiality screen in our HAP1 line of interest

- CRISPR-based Knockout
- Sampled over 21 days of culture
- **High quality data can now inform gene selection, with higher accuracy for our screens**
- **Confidence that scores represent cell behaviour expected in our screens**

> G3 (Bethesda). 2017 Aug 7;7(8):2719-2727. doi: 10.1534/g3.117.041277.

Evaluation and Design of Genome-Wide CRISPR/SpCas9 Knockout Screens

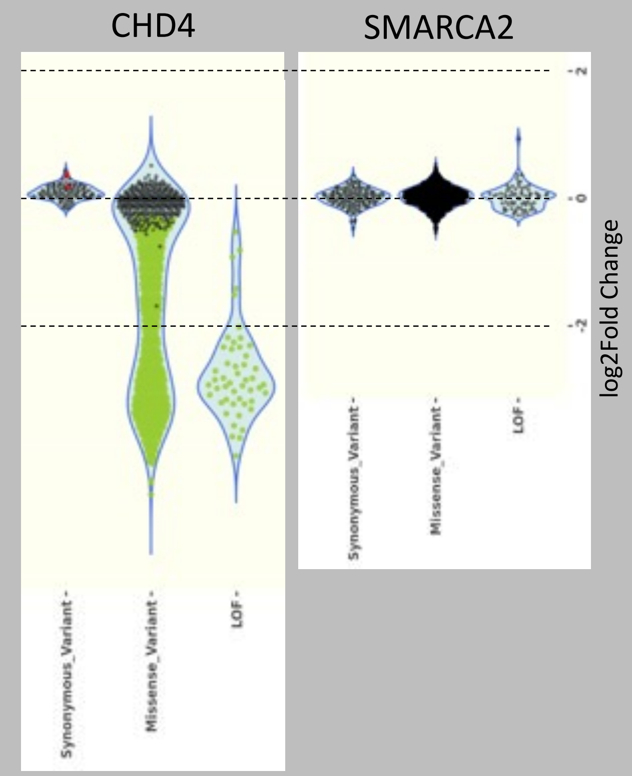
Traver Hart [1], Amy Hin Yan Tong [2], Katie Chan [2], Jolanda Van Leeuwen [2], Ashwin Seetharaman [2], Michael Aregger [2], Megha Chandrashekhar [2], Nicole Hustedt [3], Sahil Seth [4], Avery Noonan [2], Andrea Habsid [2], Olga Sizova [2], Lyudmila Nedyalkova [2], Ryan Climie [2], Leanne Tworzyanski [2], Keith Lawson [2], Maria Augusta Sartori [2], Sabriyeh Alibeh [2], David Tieu [2,5], Sanna Masud [2,5], Patricia Mero [2], Alexander Weiss [2], Kevin R Brown [2], Matej Usaj [2], Maximilian Billmann [6], Mahfuzur Rahman [6], Michael Constanzo [2], Chad L Myers [6], Brenda J Andrews [2,5,7], Charles Boone [2,5,7], Daniel Durocher [3,5], Jason Moffat [8,5,7]



black line is mean of all gRNAs

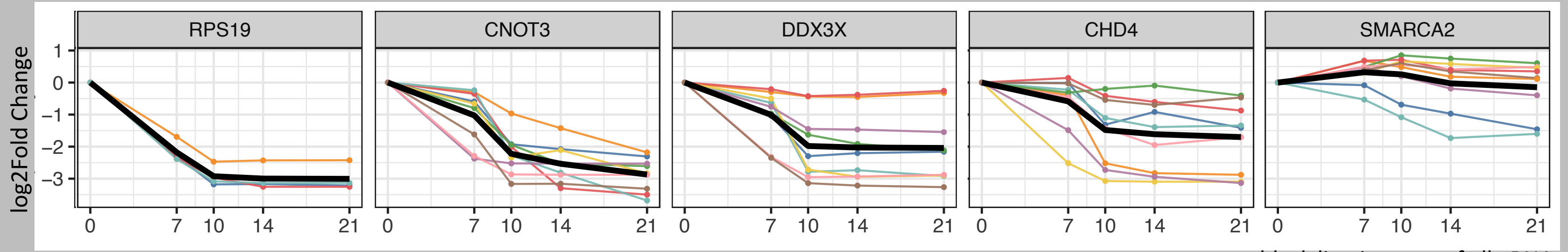
Critical factors in SGE success

Essentiality in HAP1



Accurate essentiality data is critical to planning SGE experiments

- **costs of experiments are high (~£2000/exon)**
- **failure is expensive**



black line is mean of all gRNAs

Applying SGE to Developmental Disorder Genes

Triage of Neurodevelopmental disorder genes of interest illustrates limits of HAP1-based SGE

- Neurodevelopmental disorder associated genes - 2800
- Loss of Function Mechanism – 1800 genes
    - Essentiality in HAP1 $LIG4^{KO}$
        - LFC < 1.5 at day 21 325 genes
        - LFC <2 at day 21 251 genes
        - LFC <2.5 at day 21 145 genes

        - **Therefore around 10% of NDD LOF genes may work with HAP1-SGE**
            - Great if you want to make Variant Effect maps
            - Not ideal if you have a specific interest in a specific subset of genes

Critical factors in SGE success

1. Variant sampling
We currently passage 5M cells where possible
1500 Variants in 5M cells >3000X theoretical coverage – Great!

Although HDR rates can approach 80%, 50%-60% of cells don't carry intended edits on day 4 (time = 0)

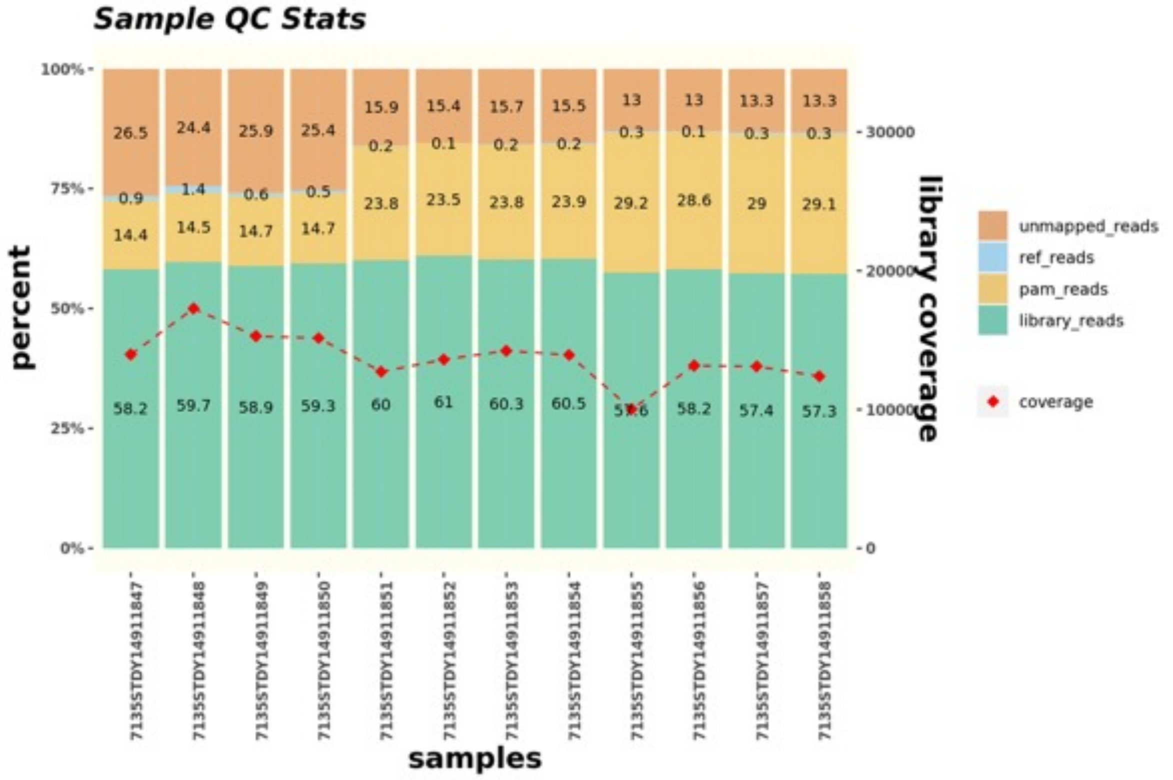
- indels
- double mutant HDR templates
- no-variant templates (wildtype or PAM edit only)
- other cloning problems

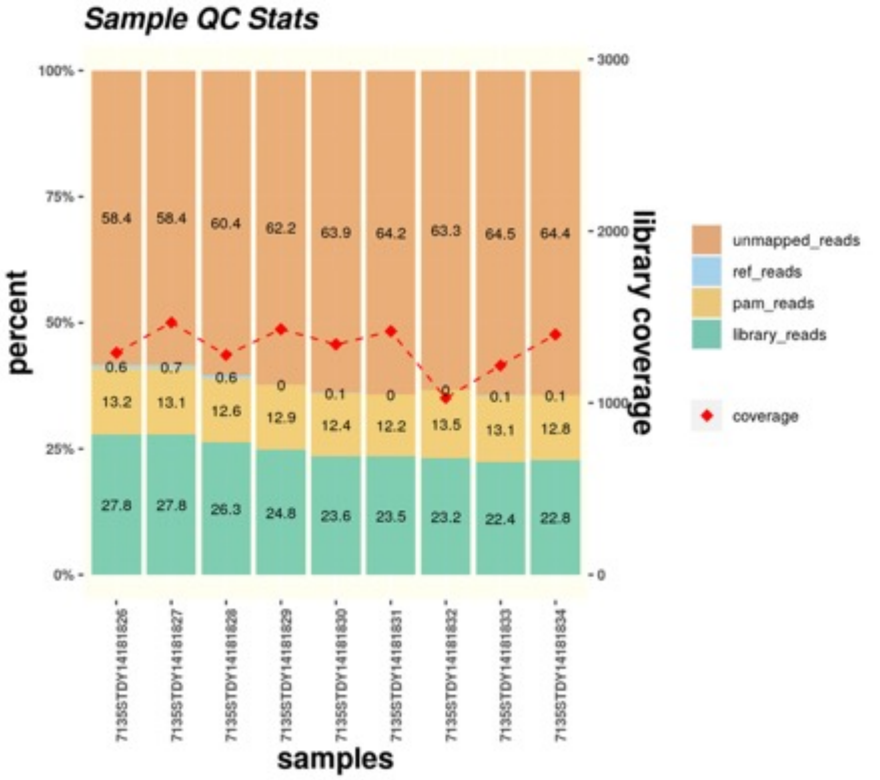
Critical factors in SGE success

1. Variant sampling
We currently passage 5M cells where possible
1500 Variants in 5M cells >3000X theoretical coverage – Great!

Although HDR rates can approach 80%, 50%-60% of cells don't carry intended edits on day 4 (time = 0)

- indels
- double mutant HDR templates
- no-variant templates (wildtype or PAM edit only)
- other cloning problems





Critical factors in SGE success

1. Variant sampling - Maximising HDR events and avoiding bottlenecking cultures
We currently passage 5M cells where possible
1500 Variants in 5M cells >3000X theoretical coverage – Great!

Although HDR rates can approach 80%, 50%-60% of cells don't carry intended edits on day 4 (time = 0)

- indels
- double mutant HDR templates
- no-variant templates (wildtype or PAM edit only)
- other cloning problems

Sources of errors

1. **Twist** 300 base oligos have **inherent error rate**: claimed 1:2000 bases = 15% of inserts (could be as high as 30%)
2. **Cloning** of 300mer inserts (saturation region) involves **PCR and Gibson assembly**
   - both introduce risk of errors in amplification, heterotypic duplex correction, and amplicon recombination
3. **HDR is dependent on**:
   - CAS9 *magic* (efficiency? timing? not recutting?)
   - Sufficiently high HDR plasmid concentration in the cell nucleus – MAXIMISING TRANSFECTION EFFICIENCY IS KEY
   - Other unknown factors

Critical factors in SGE success

1. Variant sampling
We currently passage 5M cells where possible
1500 Variants in 5M cells >3000X theoretical coverage – Great!

OK, but even 60% garbage, that leaves 40% gold dust!
- *40% intended edits could represent 1200X coverage!*

Unfortunately at least 2 factors limit this:
- Cell numbers at day 4 (time = 0) canbe lower than 5M seen in later timepoints -> Bottleneck
- Cells at day 4 are descendants of smaller pool of proliferating cells from 4 days of culture: NOT INDEPENDENT SAMPLES

Why do we need high coverage?
Even clonal cell lines have inherent variation, both epigenetic, life experience, and potential somatic differences

Assaying 1 HDR event for each variant, effectively 1 cell per variant, would be prone to noise from many biological and experimental sources

The higher the coverage (i.e. cells with same variant), the more representative the population mean will be of the true effect

Critical factors in SGE success

1. Variant sampling

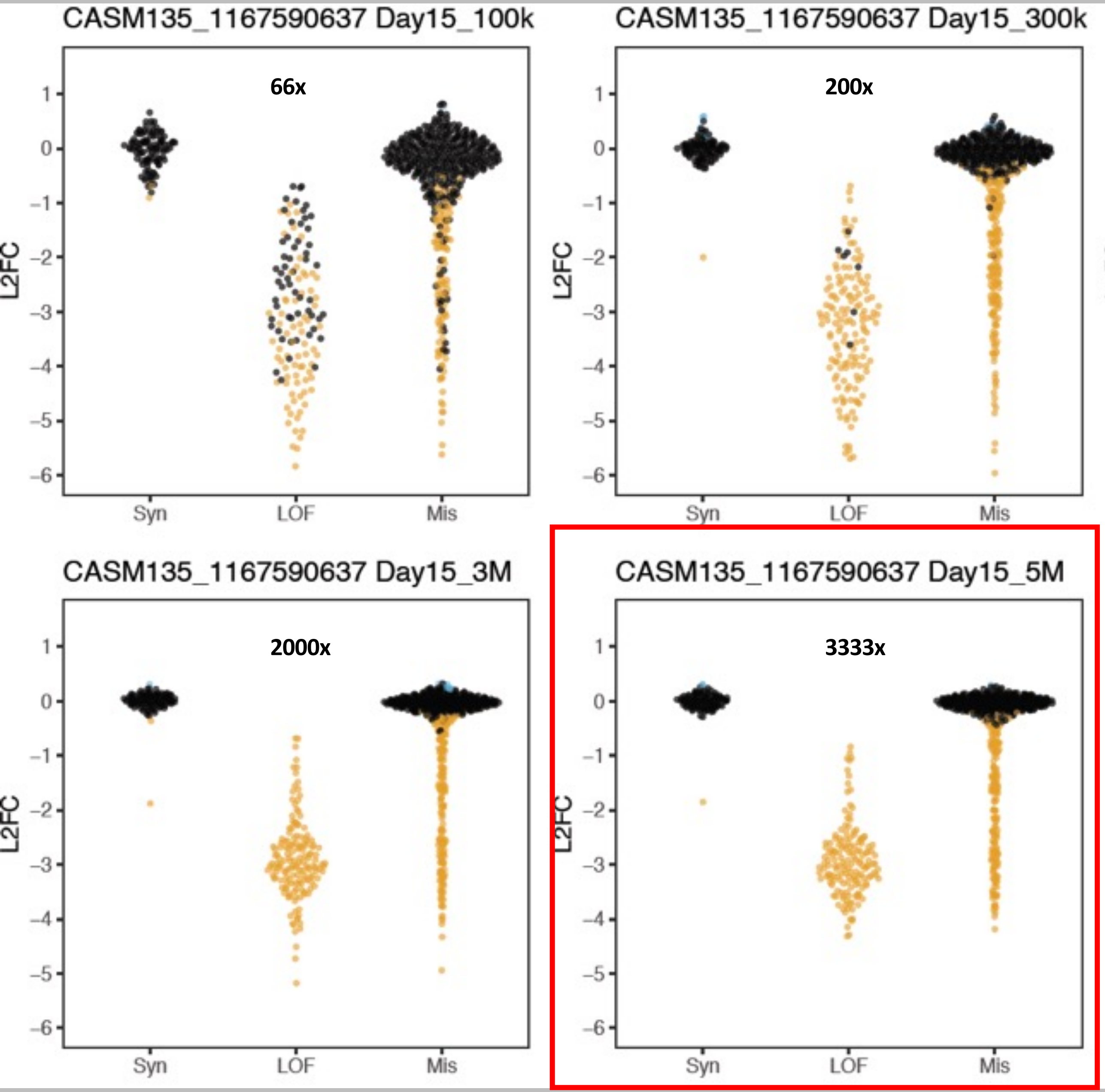
Cell scaling experiments explore empirical impact of coverage

Instead of passaging 5M cells, we passaged

- 100K cells
- 300K cells
- 600K cells
- 1M cells
- 3M cells
- 5M cells

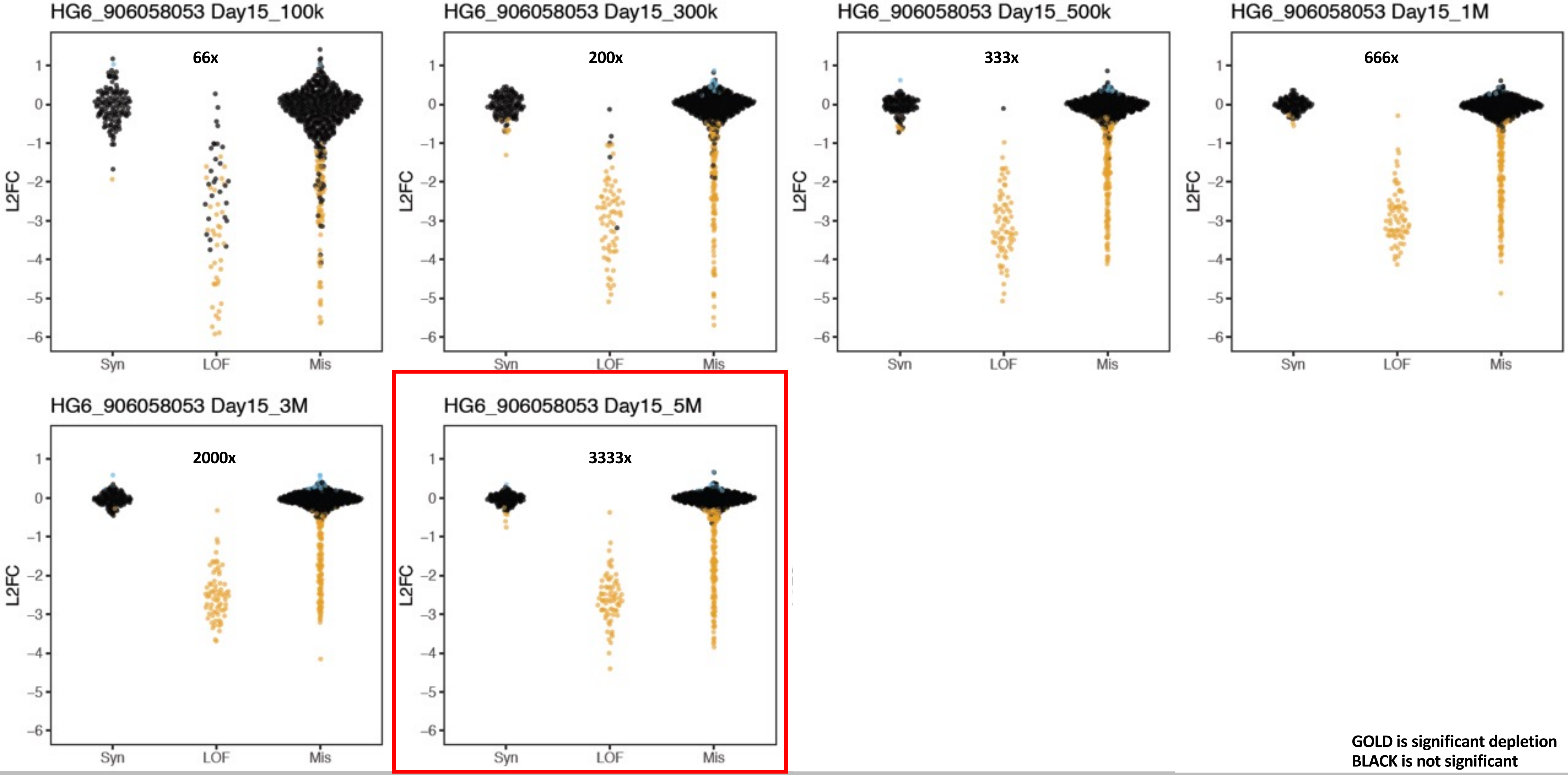
And assessed SGE performance

SGE QUALITY DROPS WITH REDUCED CELL NUMBERS



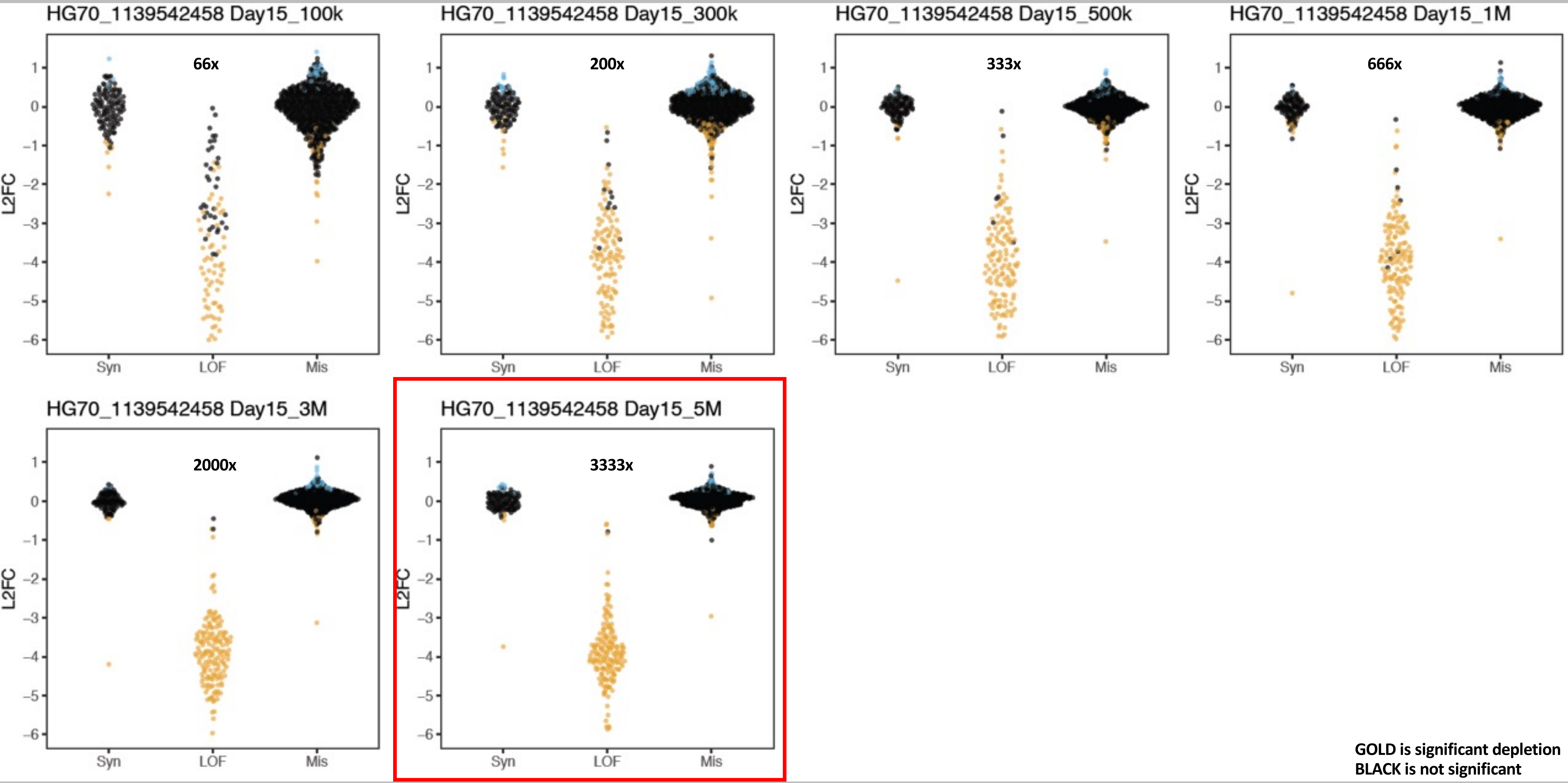
GOLD is significant depletion
BLACK is not significant

SGE QUALITY DROPS WITH REDUCED CELL NUMBERS



GOLD is significant depletion
BLACK is not significant

SGE QUALITY DROPS WITH REDUCED CELL NUMBERS



Critical factors in SGE success

1. Variant sampling

Cell scaling experiments explore empirical impact of coverage

Instead of passaging 5M cells, we passaged

- 100k cells
- 300k cells
- 600k cells
- 1M cells
- 3M cells
- 5M cells

And assessed SGE performance

CONCLUSION

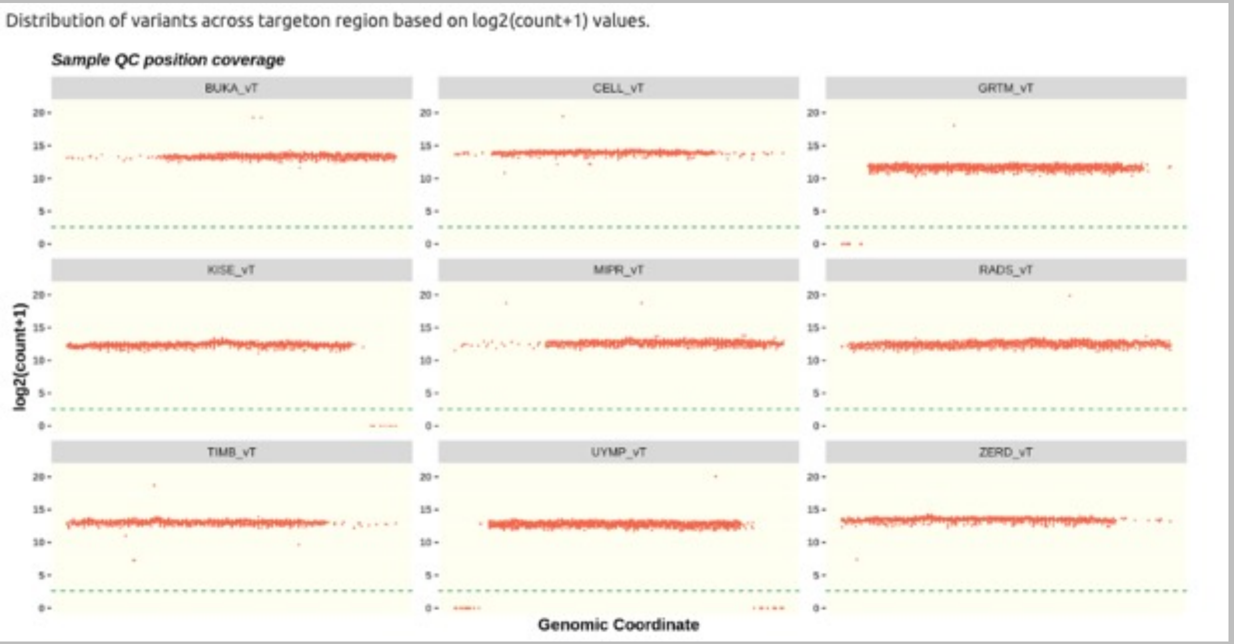
- **600K – 1M cells may be sufficient**
- **Prefer not to live life on the edge of failure: maintain 5M cells where possible**
    - **library size factors in to the numbers**

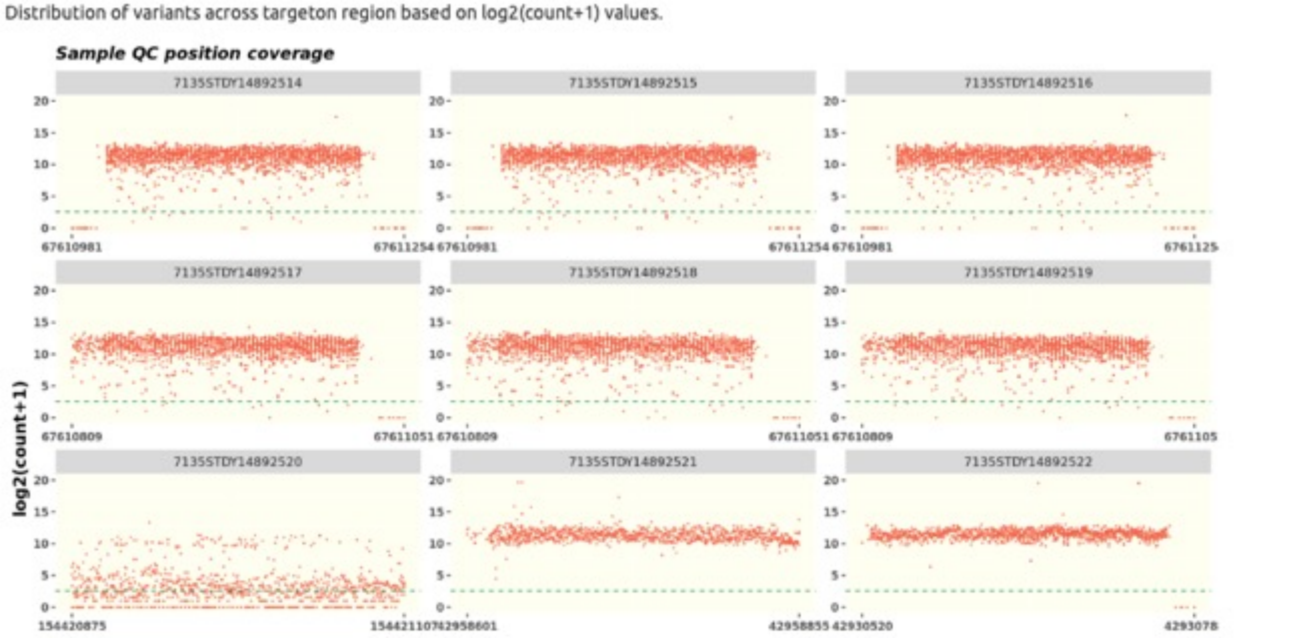
Critical factors in SGE success

1. Variant sampling

- Coverage assumes all variants represented equally
- Cloning biases and stochastic variation skew representation of variants in HDR library

PLASMID SEQUENCING AND QUANTIFICATION





Well balanced libraries have similar counts of each variant

P**oorly balanced libraries have broad distributions**

These will give varying "coverage" per variant and lower quality data

HDR plasmid QC is a critical step before attempting cell screening

Poor distribution can be caused by:

- Over amplification of HDR template oligos during PCRs
- poor coverage in cloning steps: ensure > 100X colonies per varaint in plasmid library cloning (CFU quantification)
- Over-growth or poor liquid culture of plasmid library pool

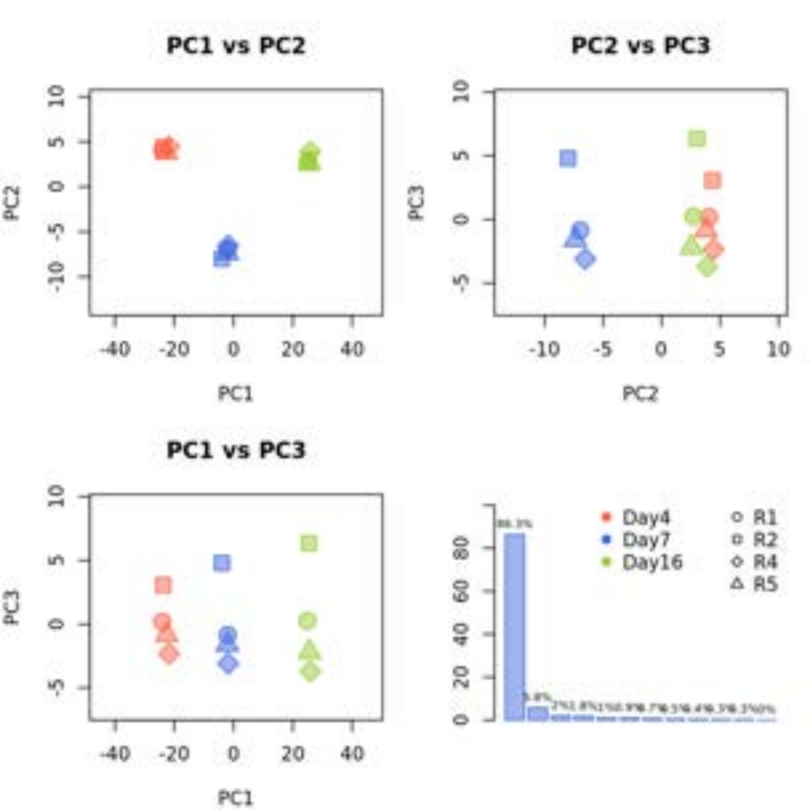
Critical factors in SGE success

2. Cell handling

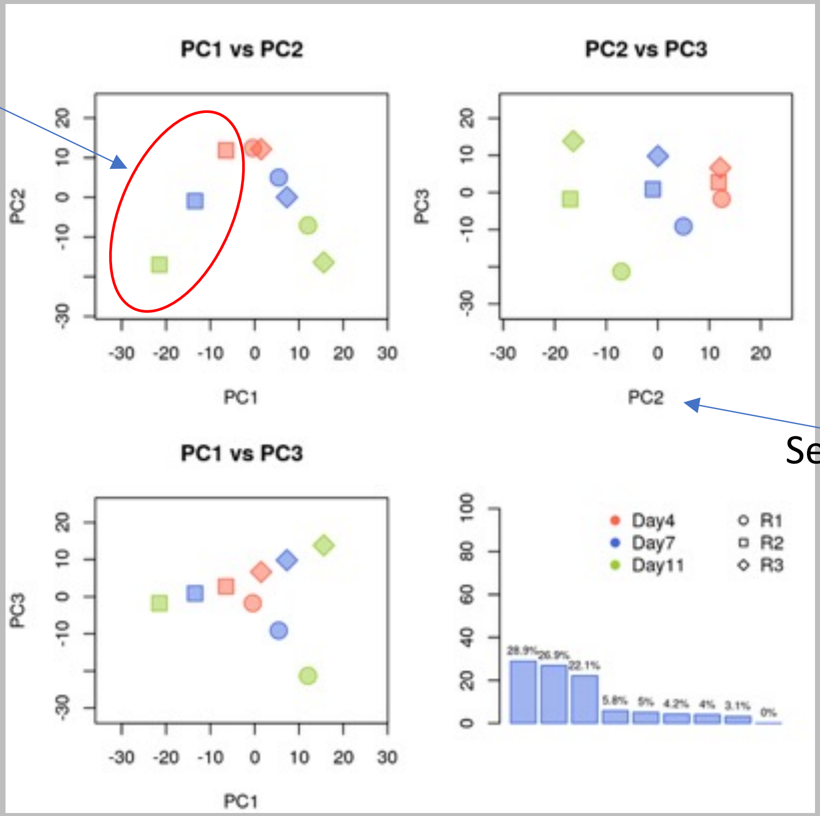
we perform 3 replicates per experiment

variation between replicates suggests uneven handling of cultures, leading to variation between reps

PLASMID SEQUENCING AND QUANTIFICATION



Replicate 1



Separates on timepoint

Happy replicates show PC1 segragating samples by time
Scree plot shows high % of variance is in PC1

Unhappy experiments show PC1 segragating samples by replicate
Scree plot shows % of variance is spread across PCs

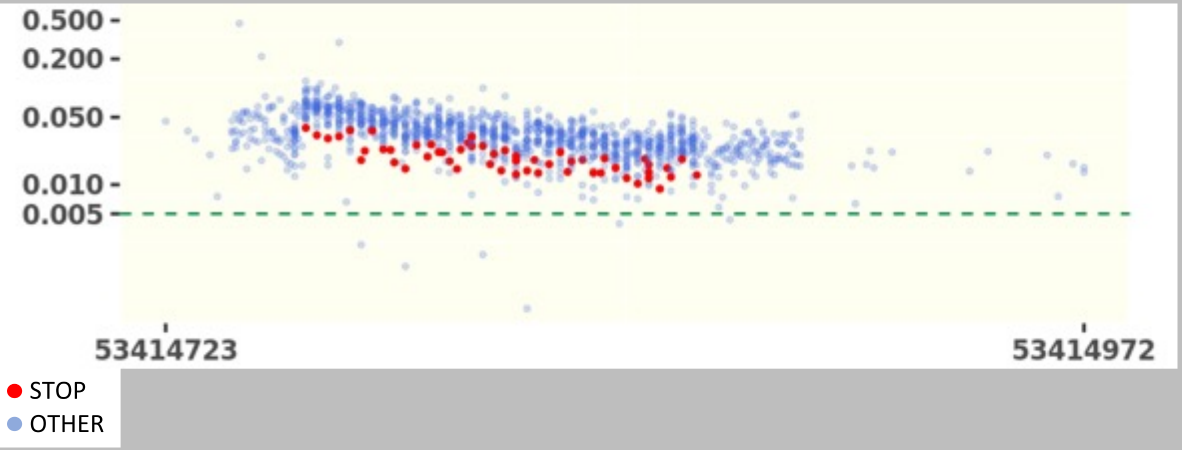
NOTE: PCA can look odd for many reasons, so don't rush blame cell handling/handler!

Critical factors in SGE success

3. Data and experimental QC

Each gene behaves differently: some a little, some a lot

SMC1A SGE: Day 4 variant representation



time = 0

Examining STOP CODONS in SMC1A revealed early depletion (at t=0)

This suggests comparing to t=0 will underestimate STOP depletion (already low)

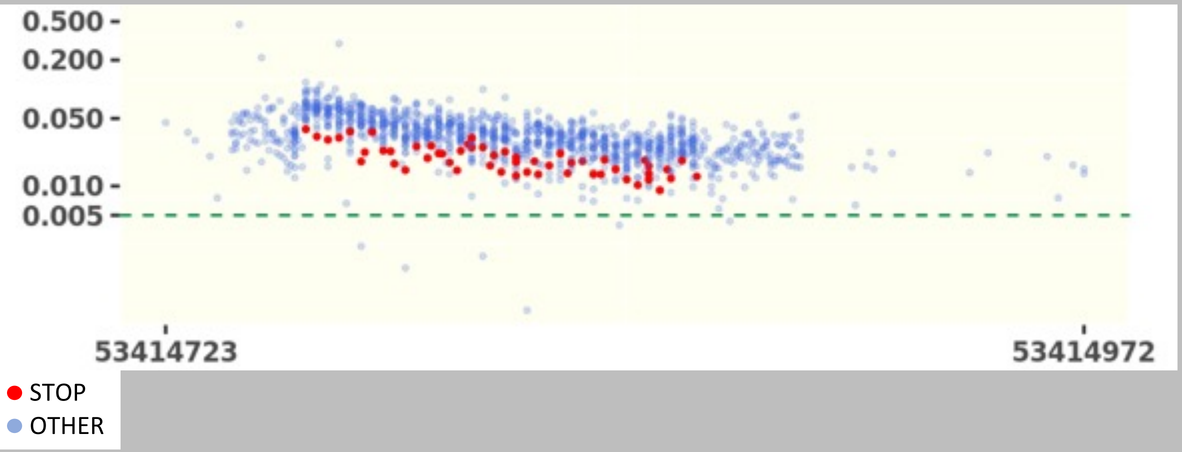
Data would be improved by using original plasmid variant representation in analysis

Critical factors in SGE success

3. Data and experimental QC

Each gene behaves differently: some a little, some a lot

SMC1A SGE: Day 4 variant representation

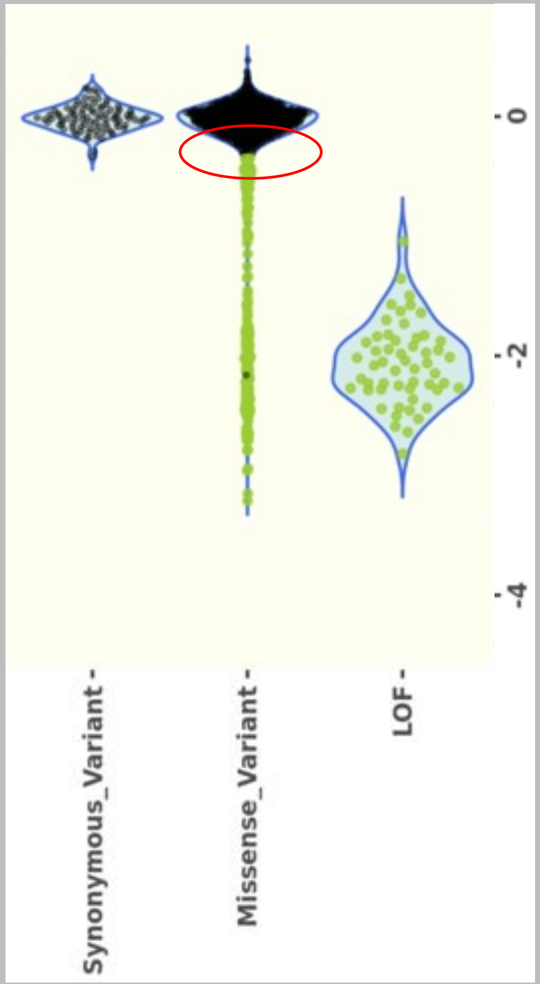


Examining STOP CODONS in SMC1A revealed early depletion (at t=0)

This suggests comparing to t=0 will underestimate STOP depletion (already low)

Data would be improved by using original plasmid variant representation in analysis

Day 15 versus day 4
SMC1A variant effects



Despite early depletion, data still *appear* good

Missense effect could have many false negatives

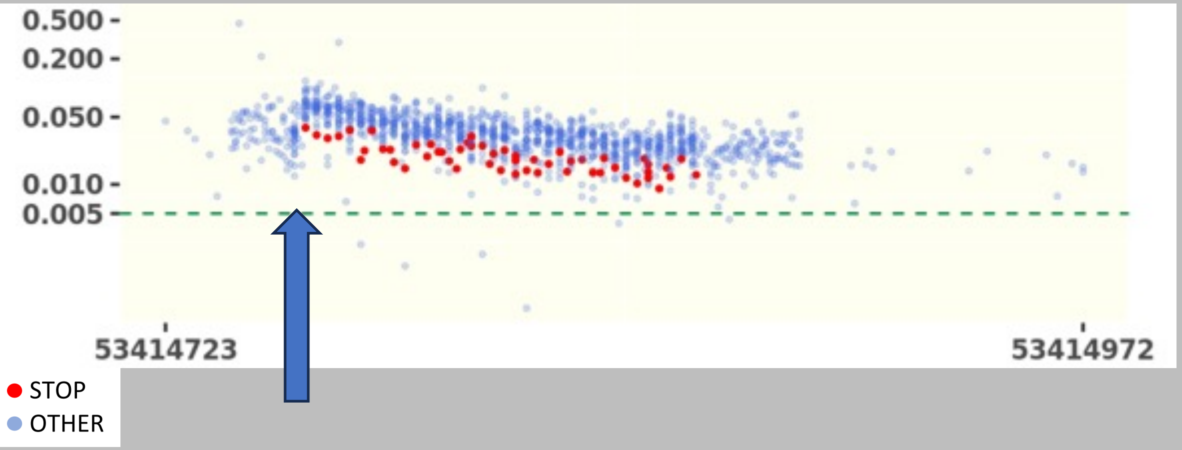
who is hiding in the red circle?

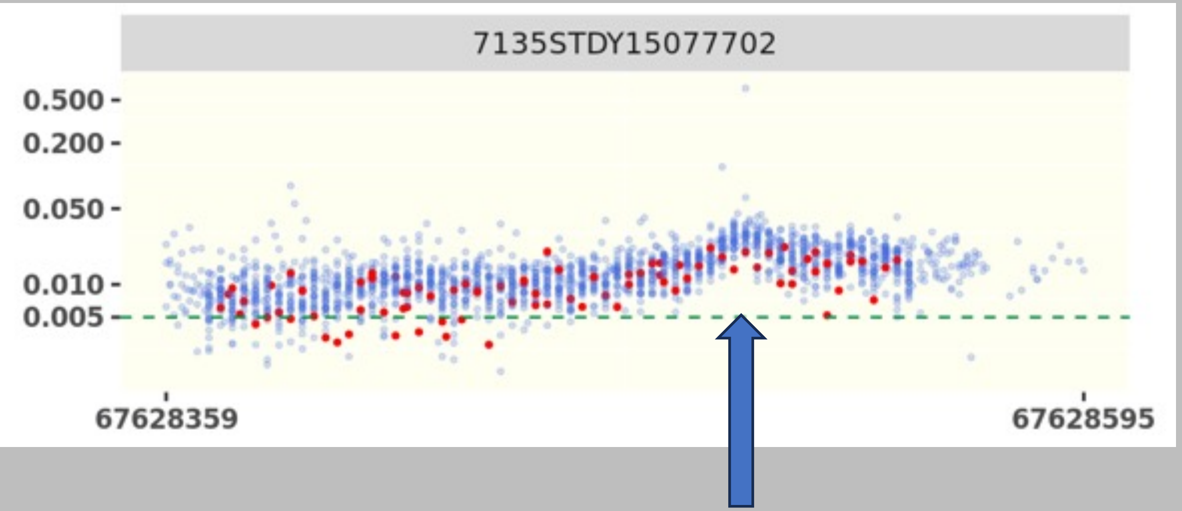
Critical factors in SGE success

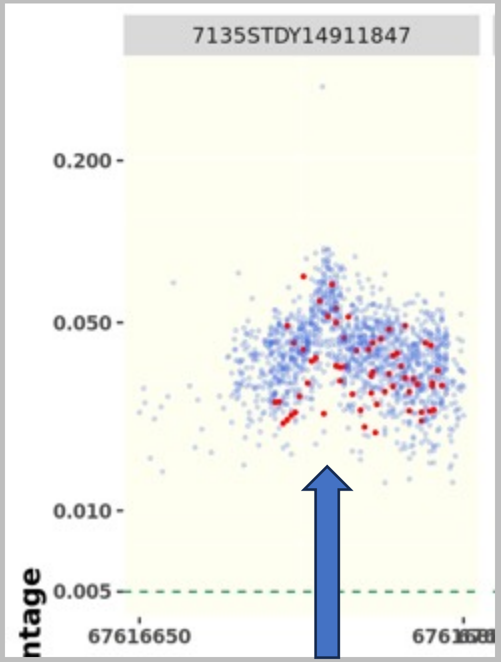
3. May need to correct for positional effects

HDR varies based on distance to CAS9 cut site, skewing representation acros variants in cell population

SMC1A SGE: Day 4 variant representation







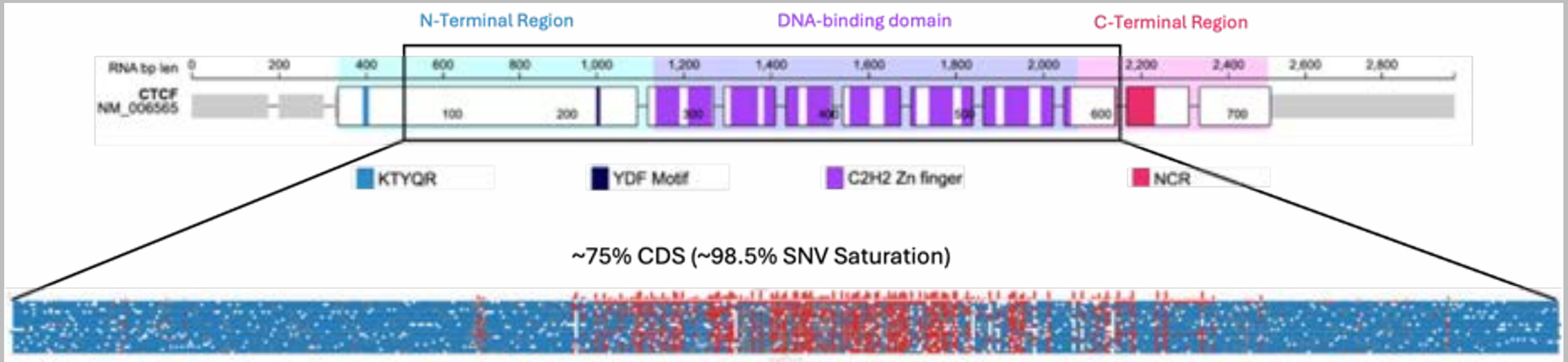
CAS9 cut sites indicated by arrow

Conclusions

Saturation Genome Editing

- High Sensitivity and Specificity for most genes we have studied
- Captures clinically important features of gene structure like splice sites, 5' UTRs, Nonsense mediated decay
  - *these are missed by many other types of MAVES*
- Despite being assayed in cancer cell line, data are proving highly specific in NDDs

- Experiments are sensitive to factors like HDR rates, library quality, cell handling, gene behaviour
- Essentiality in HAP1 limits universe of SGE–accesible genes
- Dependence on CAS9 and HDR limits SGE to a focus on individual loci
- Ideal for saturating small regions (upto 300 bases = 1 exon)
- Not well suited to broad, sparse screens
- Base Editing and (hopefully) Prime Editing are better suited to cover the "sparse-and-wide", as the technology improves.



Daniel Jaramillo-Calle
Ph.D. student

Malin Andersson
SRA

Acknowledgements

HumGen
- Deborah Plowman
- HunGen Informatics
- Vivek Iler
- Mark Thomas
- Fei Sang

Hurles group
- **Matt Hurles**
- Hong Kee Tan
- Elizabeth Radford
- Malin Andersson
- Vanessa Burns
- Rebecca Yacob
- Erwan Delage

CASM
- Dave Adams group
    - Dave Adams
    - Andrew Waters
    - Victoria Offord
    - Jamie Billington

SciOps/CellOps
- Informatics – Leah Wells
- Gene Editing Team
- Cell screening Team
- DNAP Team
- Automation team – Chris Henderson
- Stuart Fawke
- Charlotte Beaver
- Sónia Gonçlaves
- Ian Lobb
- Adam Hunter

Collaborators
- Andrew Bassett
- Sarah Cooper
- Daniel Gittermann
- Lea Starita – Wash U.
- ChatGPT
- Atlas of Variant Effects

EIF4A1 Helicase domain variant effect map